# SBC8080 (Single Board Version) Addendum

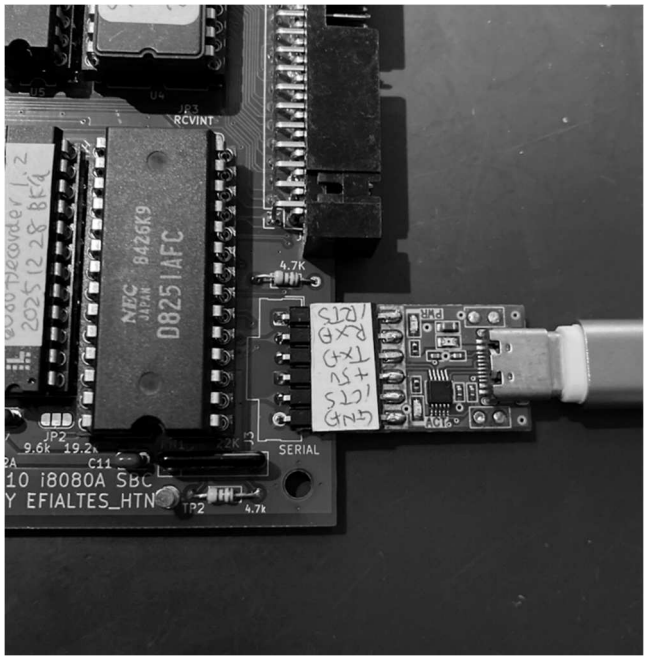
SBC8080 (Single Board Version) described here are a design derived from 2-board version by Vintagechips (Suzuki-san). This board is software compatible, and generally hardware compatible, but some intended changes mainly due to single board modification. Differences, notices and tips will be shown here. Please, refer a fine documents by Suzuki-san's for any other general topics not shown here.

This board are a part of a loose-kit. You have to orde r(or make) print-circuit board by yourself. This SBC has a high component density, so please understand that it is a step more difficult to build than the original SBC8080. Furthermore, you have to prepare EPROM and GAL programmer. China-made equipents would be a good option.

## Specifications

(1) Hardware changes

- Some components has changed. I'll show them in a parts-list chapter.
- +12V and /RESIN signals are deleted from SBCBUS. /RESIN pin are reused as INTA signal.
- The power switch has been removed.
- Since the EPROM and 8251A are directly connected to the data bus, please do not overload it.
- The connector for the serial conversion cable is oriented in the opposite direction to the SBC8080SUB board. Install it upside down compared to the SBC8080 technical documentation. The following orientation is correct.




(2) Software changes

- The address map has been modified.

  The ROM implementation area has been reduced to 16KB, from 0000H to 3FFFH. This was done to allow for the use of approximately 16KB of memory in future sub-boards.

  Also, the allocation of the 8251A in the I/O space has been narrowed to 00H to 03H. This shouldn't be a problem.
- Default serial baud rate is now set to 19,200bps. If you need 9600bps, please change a jumper shown below.

I will explain jumpers in more detail in below.

## HOW TO MAKE THIS BOARD

The parts list is attached below.

| Reference | Value | Qty | 部品仕様 |
|---|---|---|---|
| C1 | 470pF 25V | 1 | Multilayer ceramic capacitor |
| C2 | 10μF16V | 1 | Electrolytic capacitor |
| C14 | 10μF16V | 1 | polymer capacitors |
| C3 | 10pF 25V | 1 | Multilayer ceramic capacitors |
| C4 | 270μF16V | 1 | Electrolytic capacitor（for HIGH-Ripple）8mmΦ |
| C5 | 470μF 25V | 1 | polymer capacitors 8/10mmΦ |
| C6,C7,C8, C9,C10,C11, C12,C15, C16,C17 | 0.1μF 25V | 10 | Multilayer ceramic capacitors（recommend X5R/X7R） 2.54mm 巾 |
| C13 | 0.047μF 25V | 1 | Multilayer ceramic capacitors |
| D1 | 1S2076 | 1 | Signal Swithcing Diode |
| D2 | 1S4 | 1 | Schottky diode |
| D3 | LED | 1 | Any LED is OK. |
| D4,D5 | BAT43 | 2 | Schottky diode |
| J1 | Barrel_Jack_Switch | 1 | 2.1mmΦ standard DC jack |
| J2 | SBCBUS | 1 | 40-pin L-type box header |
| J3 | Serial | 1 | Single-row L-type pin header |
| L1 | 100μH PWR | 1 | 100μH（1.2A）inductor |
| R1 | 0.47 1W | 1 | Metal oxide film resistor |
| R2,R4,R5, R10,R11 | 4.7K | 5 | Carbon resistor 1/6W |
| R3 | 330 | 1 | Carbon resistor 1/6W |
| R6 | 13k 1% | 1 | Metal film resistor 1/6W |
| R7 | 1.5k 1% | 1 | Metal film resistor 1/6W |
| R8,R9 | 1K | 2 | Carbon resistor 1/6W |
| R12 | 100 | 1 | Carbon resistor 1/6W |
| R13,R14 | 27 | 2 | Carbon resistor 1/6W |
| RN1 | 22k | 1 | Resister network(4 element) |
| SW1 | RESET | 1 | Small Tactile Switch |
| TP1 | PHI2T | 1 | Optional |
| TP2 | TPGND | 1 | Optional. |
| U1 | i8080A | 1 | Intel 8080A Equivalent |
| U2 | uPD8224 | 1 | Intel 8224 Equivalent |
| U3 | i8238 | 1 | Intel 8228/8238 Equivalent |
| U4 | 27C512 | 1 | 27256/27512 Equivalent |
| U5 | HM62256BLP | 1 | 300mil Width HM62256 Equivalent |
| U6 | i8251A | 1 | Intel 8251A Equivalent |
| U7 | DEC8080A | 1 | ATF22V10 Equivalent |
| U8 | MC34063AP | 1 | MC34063A equivalent. |
| U9,U10 | 74HC244 | 2 | 74 logic。 |
| X1 | 18.432MHz | 1 | Crystal resonator HC-49U |
| S1 | 2227MC-08-03 | 1 | 8-pin IC Socket |
| S2 | 2227MC-16-03 | 1 | 16-pin IC Socket |
| S3,S4 | 2227MC-20-03 | 2 | 20-pin IC Socket |
| S5 | 2227MC-24-03 | 1 | 24-pin 300mil IC Socket |
| S6 | 2227MC-28-03 | 1 | 28-pin 300mil IC Socket |
| S7,S8,S9 | 2227MC-28-06 | 3 | 28-pin 600mil IC Socket |
| S10 | 2227MC-40-06 | 1 | 40-pin IC Socket |

・A solid polymer capacitor is recommended for C14. A 10μF 50V multilayer ceramic capacitor is also OK. While electrolytic capacitors should function without problems, the voltage output will be lower.

・Use capacitors for high ripple use for C4 and C5 (especially C5). If unsure, use solid polymer capacitors. Both should be around 220-470μF. Larger capacities will likely not fit PCB space.

・Use approximately 50-100Ω for R12. For R13-R14, use approximately 27Ω if using SN74LS424 as an 8224 compatible component. 27Ω is acceptable for Intel and NEC compatible components, but simply shorting them should also work.

・Pay attention to the package width of the U5 memory. Both 600mil and 300mil widths are commercially available. 300mil packages are sometimes sold under the name SDIP.

・U4 and U7 require programming. Please program the JED file from this website into U7. For U4, please program the file obtained from Denno Densetsu.

・TP1 and TP2 test points are for debugging. Use them is up to your preference.

・The DC-DC circuit has been modified to account for intermittent oscillation. A 47μH inductor is acceptable, but please choose one that can handle around 1A.

・1/6W resistors are assumed. 1/4W resistors may fit if forced in

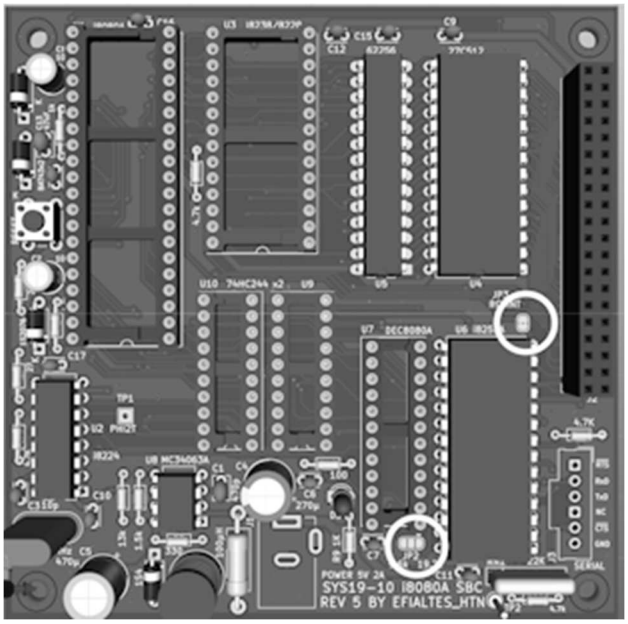
・A USB serial cable and a 5V 2A 2.1mmΦ AC adapter are also required. During debugging, I used products from Akizuki Denshi Tsusho. For the former, I used a USB cable and the "CH340E USB Serial Conversion Module Type-C" during debugging.

・Newer Akizuki Barrel_Jack_Switches appear to have plastic protrusions for position adjustment. Since this is not compatible, please break off the protrusions.

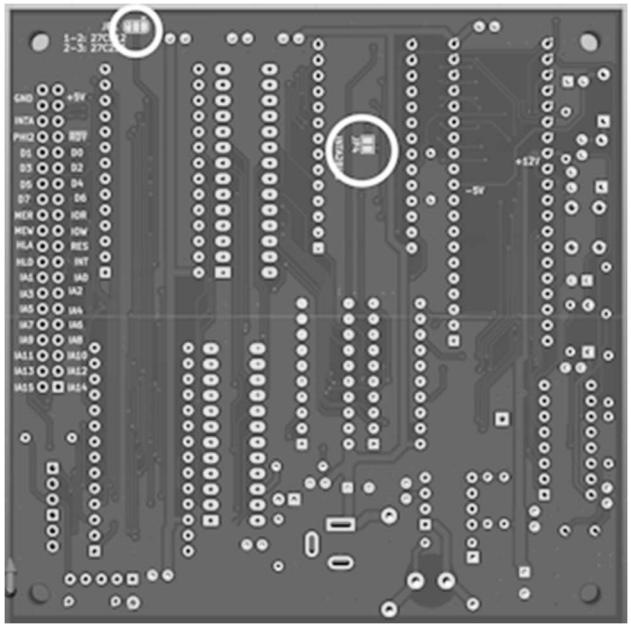
## Jumpers

This board provides several solder jumpers for configuration flexibility. The jumper locations are shown below. JP1 and JP2 are 3-pad jumpers, while JP3 and JP4 are 2-pad jumpers. The pad marked with a small triangle is pin 1. On the finished board, the default side is shorted; to change this, you must cut the default side with a knife or similar tool. JP4 only needs to be connected.

| | | |
|---|---|---|
| JP1： | Short between 1 and 2 (default). | EPROM is 27512 type. |
| | Short between 2 and 3. | EPROM is 27256 type. |
| JP2： | Short between 1 and 2 (default). | Serial speed is 19,200bps. |
| | Short between 2 and 3. | Serial speed is 9,600bps. |
| JP3： | Short (default) | Directly connects the 8251A receive interrupt to the 8080A interrupt (INT terminal). |
| | Open | The above direct connection is not performed. |
| JP4： | Open (default) | Does not connect the INTA of the 8228/38 to SBCBUS. |
| | Short | Connects the INTA of the 8228/38 to SBCBUS.<br>If you want short this jumper, remove R8 (1kΩ) in the upper left. |




Position of JP2、JP3



Position of JP1、JP4

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